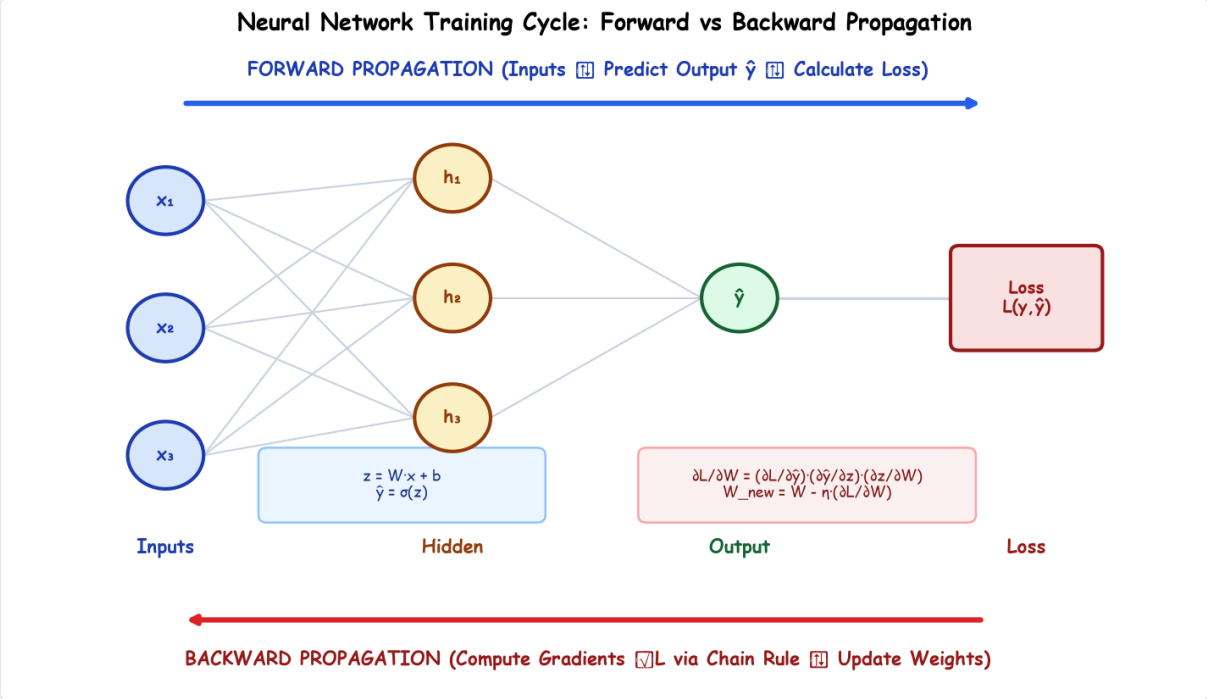
Lecture 37: Deep Learning Foundations (Part-2) — Training Neural Networks (Hinglish)

Yeh **Lecture 37: Deep Learning Foundations (Part-2)** ka master guide hai. Is lecture me humne seekha hai ki ek Neural Network asal me train kaise hota hai: Loss functions, Backpropagation, Calculus ka Chain Rule, Vanishing Gradients, ReLU ke variants, aur Modern Optimizers (Adam, RMSProp, Momentum).

Training Cycle: Forward vs Backward Pass



1. **Forward Propagation**: Inputs $\rightarrow$ Linear Sum ($z = Wx + b$) $\rightarrow$ Activation ($a = g(z)$) $\rightarrow$ Prediction $\hat{y}$ $\rightarrow$ Loss Calculation $\mathcal{L}(y, \hat{y})$.
2. **Backward Propagation**: Loss se shuru hokar Chain Rule ke zariye har layer ke weights aur biases ke gradients nikaalna.
3. **Weight Updation**:

$$W_{new} = W_{old} - \eta \cdot \frac{\partial \mathcal{L}}{\partial W}$$

$$b_{new} = b_{old} - \eta \cdot \frac{\partial \mathcal{L}}{\partial b}$$

Submodules ki List

Submodule	Topic	Mukhya Concept	Documentation
37.01	Forward & Backward Propagation	Do-engine training cycle aur mathematical flow	English | Hinglish
37.02	Loss Functions (Regression)	Loss vs Cost, MSE, MAE, Huber Loss	English | Hinglish
37.03	Loss Functions (Classification)	Binary Cross-Entropy, Softmax, Categorical Cross-Entropy	English | Hinglish
37.04	Weight & Bias Updation	Gradient descent formula aur slope direction analysis	English | Hinglish
37.05	Chain Rule of Derivatives	Multi-variable chain rule se backpropagation ka proof	English | Hinglish
37.06	Vanishing Gradient Problem	Sigmoid derivative 0.25 limit aur gradient decay	English | Hinglish
37.07	ReLU & its Variants	Dying ReLU problem, Leaky ReLU, PReLU, ELU	English | Hinglish
37.08	Batch vs Iteration vs Epoch	Batch size, iterations per epoch aur memory balance	English | Hinglish
37.09	Classical Optimizers	Batch GD, Stochastic GD, Mini-batch GD comparison	English | Hinglish
37.10	Modern Variants of GD	Momentum, RMSProp, Adam optimizer	English | Hinglish

| **Practice** | Industrial Practice Suite | 15 technical problems aur step-by-step solutions | Questions | Solutions |\n